Proceedings of 2024 International Conference on Cognitive Robotics and Intelligent Systems (ICC - ROBINS 2024)
IEEE Xplore Part Number: CFP24UD1-ART; ISBN: 979-8-3503-7274-8

2024 International Conference on Cognitive Robotics and Intelligent Systems (ICC - ROBINS) | 979-8-3503-7274-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICC-ROBINS60238.2024.10533954

Weather-Integrated Recommendations for Food Choices with SHAP Explanation

Nikunj Jain
Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse21138@bl.students.amrita.edu

Jawed Hawari
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse21073@bl.students.amrita.edu

Priyanshu Jha
Department of Computer Science and Engineering,
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
bl.en.u4cse21165@bl.students.amrita.edu

K Dinesh Kumar
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India
kk_dinesh@blr.amrita.edu

Abstract*— In the context of today's rapidly evolving technological landscape and the urbanization of Tier 1 and Tier 2 cities, cloud kitchens have emerged as pivotal players, offering quick and convenient meals without the need for dine-in facilities. The demand prediction for the food has become quite tough and could be considered as one of the significant challenges. Cloud kitchens face challenges in managing stock due to the absence of in-house diners, potentially resulting in wasted food. Moreover, adverse weather conditions often hinder the seamless delivery of online. To address these challenges, this project employs machine learning techniques to predict weather patterns and correlate them with food preferences according to that weather. The key algorithms found to deal with this problem were Random Forest, kNN, Decision tree and Naïve Bayes and Support Vector Machine. This serves the first half of the statement and then for recommendation a suitable system was developed. The pivotal role of the system is to reduce the food wastages for cloud kitchens by fostering adaptive strategies for efficient decision making. The key inputs from the market were found to be that 30-40% of the predicted sales in food industry doesn't meet due to various reasons. The ultimate goal is to empower cloud kitchen operations, ensuring adaptive and informed decision-making in response to changing weather dynamics. The union of both of these are explained using Interpretability tool SHAP for a convenient study.

Keywords: **Cloud kitchens, Decision Tree, Random Forest, Recommendation system, SHAP, Weather pattern**

I. INTRODUCTION

Weather predictions serve as a critical determinant in various sectors, playing an indispensable role in strategic decision-making, planning, and risk management. For the food industry, especially cloud kitchens and restaurants, understanding and predicting weather patterns is top. Unforeseen weather fluctuations significantly impact customer behavior and preferences, leading to variations in demand and sales. Predicting the weather can help businesses plan better for changes in customer demand and manage their resources efficiently. By using advanced weather forecasts and smart computer programs, companies can adjust their plans based on expected weather forecasts. This helps them avoid problems caused by sudden shifts in customer behavior due to weather changes. This active approach is especially helpful for cloud kitchens and restaurants. It allows them to organize their work better, use their resources wisely, and reduce food waste caused by unpredictable weather. With technology getting better, there are different ways to predict the weather. The study in this article explores at various methods to find the most accurate and efficient ones.

Throwing away food isn't just about losing meals and wasting it. It also means losing a lot of money for businesses and causing big problems for the environment. Businesses lose money when they can't sell the food they've prepared, and it costs money to get rid of the extra food. Plus, making food uses up a lot of resources like water and energy, and when food gets wasted, it's like wasting all those resources, too. Cloud kitchens, which are becoming more popular, have a hard time managing how much food they need and reducing waste. Because they don't have physical customers coming in, things like bad weather affecting food deliveries can make it even harder for them. According to a report by Food and Agriculture Organization (FAO), for each food produced, one third of it goes to mere waste which eventually challenges the sustainability of the planet.

When weather forecasts and food businesses work together, it can really help tackle problems. For cloud kitchens, using these weather predictions in their planning can make a big difference. When they use smart systems that learn from the weather forecasts, they can make food better with less waste and adjust their plans. This integration between weather predictions and food businesses helps these kitchens work better, matching the changing weather and what customers want, making things more efficient and eco-friendly.

II. RELATED WORK

In [1], researchers have explored how food supply chains are changing. The author states that there are several digital ways to handle the supply chains. In [2], the author has designed a complex relationship between food distribution and consumption. The author's research provided valuable insights for policymakers, offering empirically constructed policy models to identify potential behavioral consequences. In [3], the proposed method has integrated harnessing

Proceedings of 2024 International Conference on Cognitive Robotics and Intelligent Systems (ICC - ROBINS 2024)
IEEE Xplore Part Number: CFP24UD1-ART; ISBN: 979-8-3503-7274-8

technology to create an efficient system for managing food wastes and predicting the right amount of food that restaurants should prepare by using historical data by ultimately minimizing the wastage. In [4], authors deal with weather forecasting. Many modern weather models were taken into account with different factors such as temperature, humidity, wind, and precipitation, leading to more predictions that are precise.

In [5], authors presented as case study to show how the Indian market for cloud kitchens is rapidly growing, contributing significantly to the booming online food delivery sector. The authors in [6] have urged U.S. restaurant leaders to reduce significant food waste expenses by integrating cloud kitchen. This research is conducted through an extensive review of existing literature to address a critical gap in current knowledge. The research proposed in [7], explored the impact of product recommendations on consumer buying behavior, particularly in the form of reviews and rankings. The paper goes a step further by introducing a recommendation system that focuses on personalized dietary advice. By employing deep learning and genetic algorithms, this system provides tailored dietary recommendations based on user experiences with specific products. In [8], authors examine the application of information technology, specifically data mining, data retrieval, and AI in the development of weather and climate prediction models and present studies with this concept of equality.

The approach outlined in [9] minimize "food loss" by introducing a system suggesting menus and recipes based on available home food items and those nearing expiry. Meanwhile, in [10], the study introduced a model for suggesting dishes based on user preferences and consumption history. This involved crafting a classification model to generate a roster of recommended dishes from the user profile. Subsequently, a temporal model utilized dish lists and eating history as inputs to enhance the variety of suggested dishes. To enhance accuracy, a set of ingredients was curated to refine the model's performance.

The authors in [11] proposed a new model to predict weather for implementing a smart IoT based irrigation-based system, this study portrays the potential of using techniques to solve real life problems. The researchers in [12] proposed an automated Irrigation system using the weather prediction to mitigate the misuse of water resources. In the study conducted by authors in [13] proposed a recommendation system which utilizes the Weather patterns using Logistic regression and big data analytics tools. The researchers in [14] advises how a recommendation-based system advances the business and ease of making complex things easier. The researchers in their study [15] suggest weather prediction to be a fascinating domain ahead and weather prediction is the estimation of the future. The study proposed in [16] delves into the building a recommendation system which enhances the user's interpretability and shopping experience by utilizing the fashion recommendation using Neighbour Page Rank Algorithm.

The studies suggest that there are multiple challenges faced by the food industry as they generate a heavy food wastage due to multiple circumstances. Current studies underscore the potential of machine learning techniques such as Random Forest, kNN, Decision Tree, Naïve Bayes, and Support Vector Machine in addressing the weather prediction. However, future research could delve deeper into the optimization of these algorithms to enhance the predictive accuracy and efficiency. While there is vast research in weather prediction the integration of it with other food industry remains underdeveloped. The proposed system integrated weather prediction and food industry for reducing the wastage and improving the business.

III. PROPOSED METHODOLOGY

This section presents a method that merges advanced machine learning techniques with comprehensive data analysis methods to address the challenges of food wastage reduction and demand prediction in the cloud kitchen sector. The proposed methodology incorporates various steps of weather predictions using advanced machine learning algorithms such as Random Forest, Decision Tree, Naïve Bayes, and SHAP. Subsequently, this weather data is meticulously integrated with food industry trends and consumer preferences.

This innovative system leverages the power of current weather data, machine learning, and recommendation engines to deliver personalized meal recommendations based on forecasted weather conditions for Weather user-supplied information, preliminary data processing, product selection, weather forecast modeling, integration and recommendation process mix well, enabling rational decisions to be made based on expected weather conditions on. The generated food recommendations are presented to the user, empowering them to make informed decisions about their dining choices based on the predicted weather. The Fig. 1 explains in detail about the workflow proposed. It depicts the step-by-step procedure which are followed in the workflow.

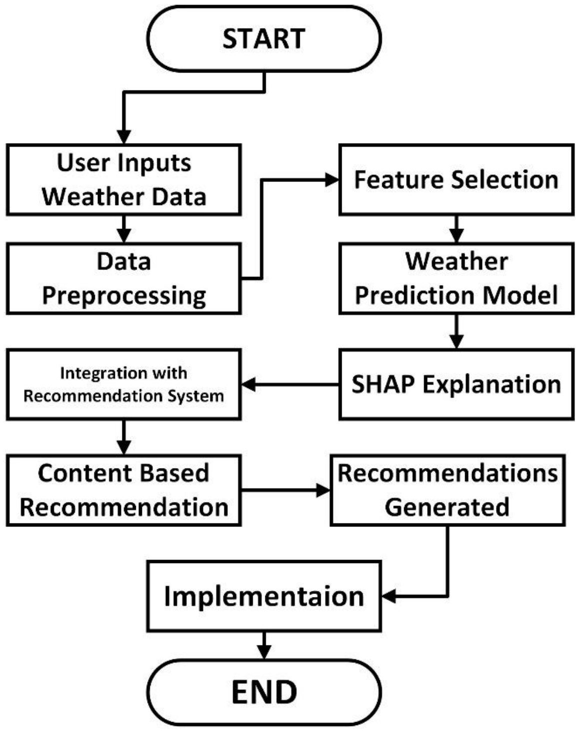

Fig. 1. Workflow of proposed methodology

A. Data Collection and Preprocessing

The dataset utilized in this study involves Machine Learning (ML) techniques to identify the weather and then recommendation based on a category. The weather dataset included Date, Time, Temperature (F), Dew Point (F), Humidity %, Wind, Wind Speed (mph), Wind Gust, Pressure, and Precip Condition. Fig. 2 presents the raw dataset without requiring any preprocessing steps. The dataset from [17] was used with location specific to Bangalore, India.

	Date	Time	Temperature F	Dew Point F	Humidity %	Wind	Wind Speed mph	Wind Gust mph	Pressure in	Precip.	Condition
2853	NaT	01:10:00	70	70	100	NW	9	0	26.75	0.0 in	Rain
37	NaT	10:00:00	88	64	46	SW	10	0	26.75	0.0 in	Partly Cloudy
603	2015-02-12	18:00:00	75	75	100	E	6	0	26.97	0.0 in	Fair
2995	NaT	05:00:00	77	66	69	NE	10	0	26.94	0.0 in	Fair
2501	NaT	02:00:00	72	72	100	CALM	0	0	26.86	0.0 in	Fog

Fig. 2. Weather dataset

The data preprocessing steps are done for removing the unnecessary features for the study. The dataset consisted of 3305 rows and 11 features. As a preprocessing step the temperature and dew point were standardized to SI unit i.e. Celsius to facilitate uniformity in calculation and analysis. As a preprocessing step, Under the "Condition", classifications were brought to 5 from 22. The 5 classifications found suitable for the study are "Partly Cloudy", "Fair", "Rain", "Fog", "Mostly Cloudy". The label encoder was utilized to encode the classes from 0 to 4 respectively.

Food categories were chosen and based on five weather classifications, corresponding dishes were included. Each entry was then supplemented with consumer behavior data and their opinions for further discussion. Upon selecting specific food categories, a variety of dishes was allocated according to five distinct weather classifications. Subsequently, each entry underwent consumer behavior insights and their individual opinions, promoting comprehensive discussions and analysis. No preprocessing steps were performed for the performance. The dataset consists of 190 rows and three features. The Fig. 3 portrays the light on the sample of dataset.

	Food items	Weather	Khana
168	Snacks Sweets	Fair	Jalebi, Gulab Jamun, Rasgulla, Barfi, Ladoo
21	Bakery Snacks	Partly Cloudy	Croissant, Puff Pastry, Danish Pastry, Samosa,...
61	Desserts	Fair	Sheer Yakh, Jelabi, Firnee, Falooda, Baklava, ...
94	Indian Fast Food	Foggy	Pav Bhaji, Chole Bhature, Samosas, Aloo Tikki,...
57	Continental	Fair	Pizza, Pasta, Burger, Steak, Lasagna, Croissan...

Fig. 3. Food dataset

B. Data Visualization and Outliers Identification

Data visualization is a pivotal aspect of data analysis, enabling individuals to comprehend complex datasets and derive meaningful insights. It involves the graphical representation of data to explore trends, patterns, and relationships within the dataset. Outliers are data points that significantly deviate from the majority of observations, potentially skewing statistical analyses and models.

To know the dataset effectively, the data visualization plays a pivotal role in effective study. The Fig. 4 portrays the relationship between Temperature and humidity through scatter plot not based on Time based sequence

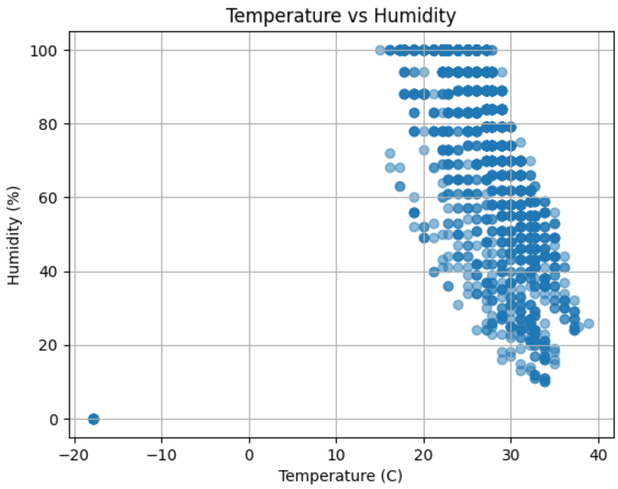

Fig. 4. Relation between Temperature and Humidity

The figure labeled as Fig. 5 illustrates the relationship between Temperature and Dew Point within the dataset. This graphical representation visually showcases how changes in temperature relate to the dew point values.

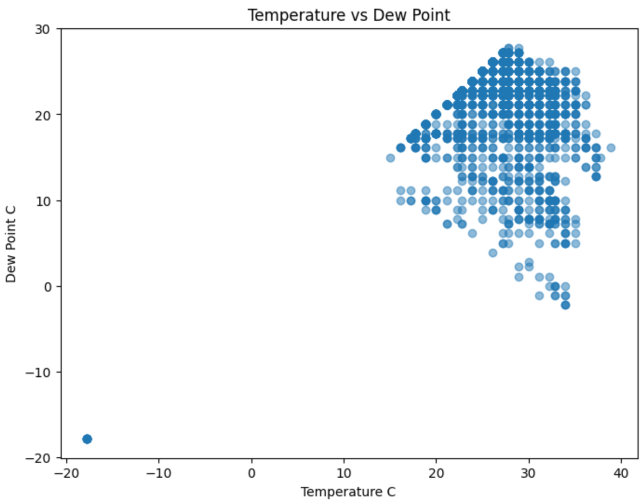

Fig. 5. Relation between Temperature and Dew Point

Detecting outliers is essential as they could indicate data collection errors, anomalies within the data, or rare events that necessitate further investigation or specialized treatment during analysis. Upon identification, these outliers are carefully examined, corrected, or addressed based on their impact and relevance to the dataset. The outliers identified are presented in Fig. 6.

Proceedings of 2024 International Conference on Cognitive Robotics and Intelligent Systems (ICC - ROBINS 2024)
IEEE Xplore Part Number: CFP24UD1-ART; ISBN: 979-8-3503-7274-8

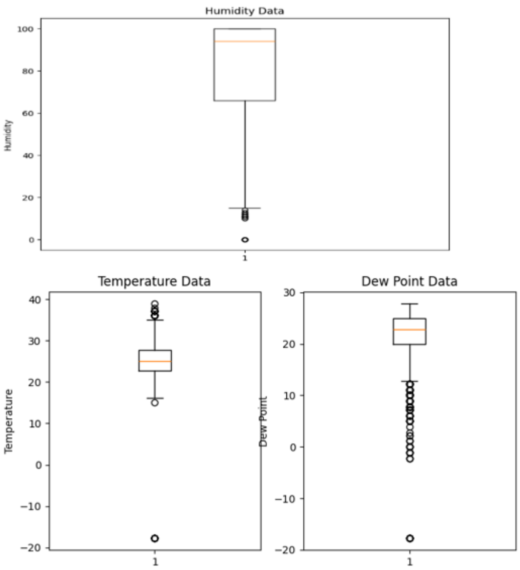

Fig. 6. Outliers Identifications

C. *Machine Learning Techniques*

The study employs a two-fold approach for the problem statement. The goal is to predict weather patterns and recommend suitable food choices based on these predictions.

1. *Random Forest:* Random forest was chosen to handle the complex data and to offer predictions in classification tasks. It has been adapted to predict weather patterns based on historical data
2. *K-Nearest Neighbors (kNN):* To utilize the proximity of neighboring points to predict weather conditions.
3. *Decision Tree:* The reason behind choosing this algorithm was its interpretability and adjust data relationship. It helps in constructing a tree to identify weather patterns and classification
4. *Naïve Bayes:* Naïve bayes was selected to utilize the concepts of conditional probability to create independence among features
5. *Support Vector Machines (SVM):* This helps in classification and handles quite complex data in simplified manner for performing tasks

Each algorithm selected went through a training phase and validation using the dataset to know its predictive performance. The proposed work evaluated on how accurate, precise, and reliable these models were in guessing weather patterns. Hereafter, the predicted weathers were then employed to recommend appropriate food categories for the cloud kitchens keeping aim to minimize the food wastage occurring due to unpredictable weather patterns.

D. *Feature Selection*

Feature selection is one of the most important parts in implementation of any Machine learning Applications. In the proposed work a consideration of selection of only important features which are necessary for the model training were selected. Various weather related and food related data points were considered to keep intact. These included Temperature, Dew Point , Wind Speed , Wind Gust, Pressure. The logic behind selection of feature was forwarded with correlation between both the components.

E. *Interpretability Tool*

Shapely additive explanations [SHAP] has been utilized as descriptive metric for knowing the contribution of various features. SHAP helps in splitting up the model's output and providing insights. With utilization of SHAP the primary aim is to ensure the transparency identify which feature has more influential factor to utilize in decision making. The SHAP Values are represented in the Fig. 7 and Fig. 8 respectively.

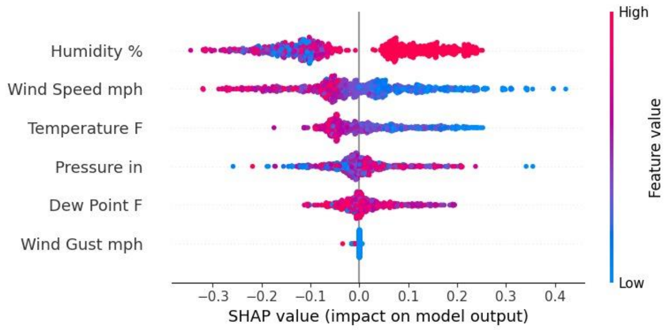

Fig. 7. SHAP Value on Features

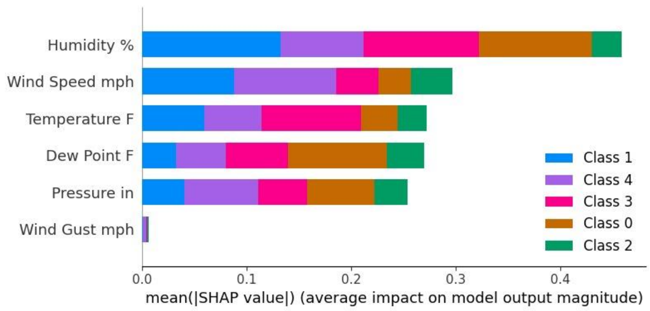

Fig. 8. Mean SHAP Value on Features

The individual SHAP value provides insight into how significantly a particular feature influenced the prediction. Each variable is arranged according to its global feature importance, with the initial variable being the most critical and the final one considered the least important. Essentially, SHAP offers a dual perspective, allowing us to observe the global contribution through feature importance and the local feature contribution for every instance.

F. *Food Recommendation*

Leveraging the insights derived from weather prediction, this system incorporates a sophisticated algorithm that correlates weather patterns with food preferences. Based on the predicted weather conditions, the system recommends a curated list of food categories and items best suited for the anticipated climate. The main goal of this recommendation system is to streamline cloud kitchen operations by dynamically adapting food offerings in response to changing weather dynamics, thereby reducing food wastage and meeting customer preferences more effectively.

In this section, we delve into the comprehensive analysis and interpretation of the outcomes derived from the implementation phase. This segment focuses on assessing the performance and efficacy of five distinct machine learning algorithms employed for weather prediction and food recommendation tasks. Each algorithm's accuracy, precision, recall, and F1-score are meticulously evaluated, providing a comprehensive understanding of their individual strengths and limitations. Additionally, the discussion encompasses the methodology adopted for handling outliers within the dataset, highlighting the impact of outlier treatment on the overall predictive model's performance.

After identifying outliers in Fig. 6 within the dataset, the parallel phase involved evaluating the performance of the ML models. The removal of outliers aimed to refine the accuracy and reliability of predictions derived from the algorithms. Upon the exclusion of outliers, a notable enhancement in the overall performance of the models was observed. The refined dataset led to more robust and accurate predictions, aligning more closely with the expected outcomes. The revised version of dataset with removal of outliers is depicted in Fig. 9.

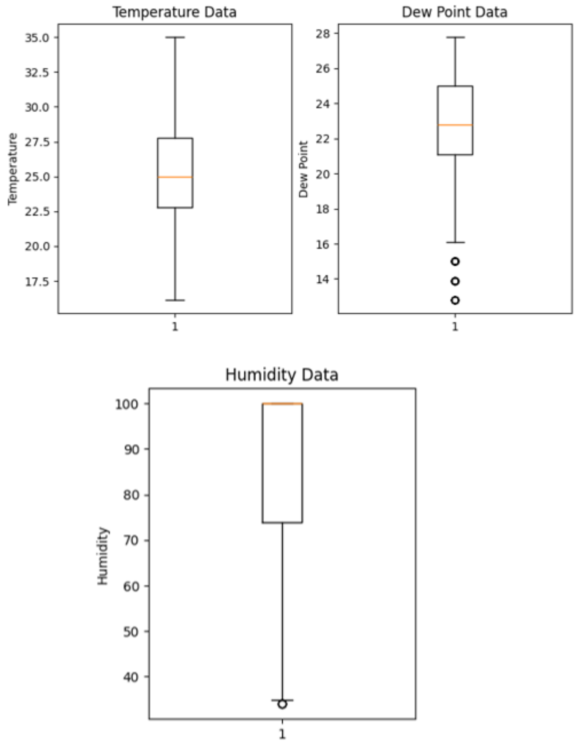

Fig. 9. Dataset without Outliers

Implementing 5 different algorithms provided with various metrics for each. The metrics like Accuracy, Precision, Recall and F1-Score for each of the following are tabulated in Table 1.

Table. 1. Metrics of 5 different Algorithms

	k-NN	Decision Tree	Random Forest	Naïve Bayes	SVM
Accuracy	0.65	0.72	0.59	0.57	0.58
Precision	0.59	0.72	0.59	0.63	0.58
Recall	0.59	0.72	0.61	0.63	0.58
F1-Score	0.62	0.72	0.59	0.63	0.58

The 5 various algorithms had been selected due to their various adaptability. Considering a fact in mind that choosing a single algorithm's result could lead incur loses for business. So for proposed work, a maximum predicted value is considered. A window is created where user can enter the weather details and the results are displayed. The user window allowed the user to enter the number of category present in their restaurant. Upon the need the user can enter the detail and get predictions/recommendations according to weather. The basic work flow in simple terms with integration of user interactive has been portrayed in Fig. 10.

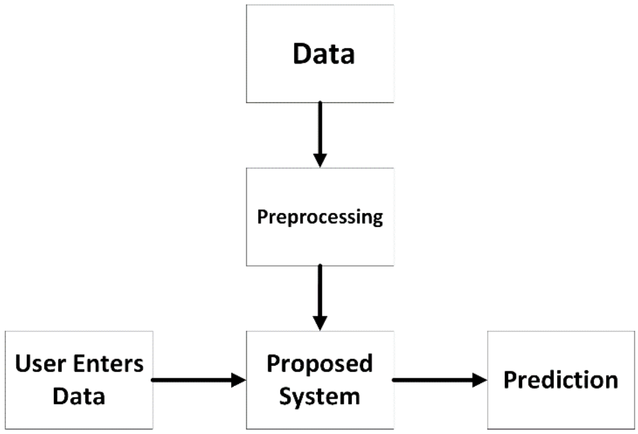

Fig. 10. User Interactive Workflow of System

The sample input and output windows are portrayed in the subsequent figures.

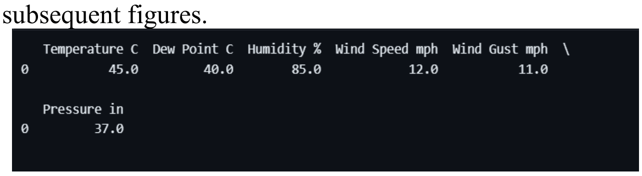

Fig. 11. Sample Input from the user end

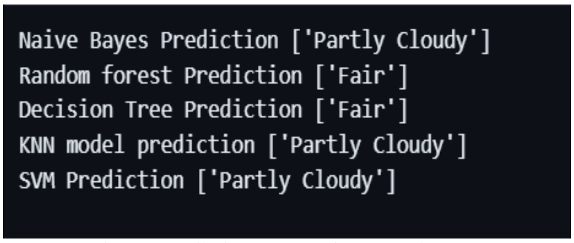

Fig. 12. Prediction generated on Sample Input

From the Fig. 11 and Fig. 12 the user can derive to a conclusion where a user on entering the details can have a prediction and conclude to be "Partly Cloudy" in this case. The further steps include the generation of Food categories by identifying highest number of counts of predicted value. Food categories do include various food varieties across the globe.

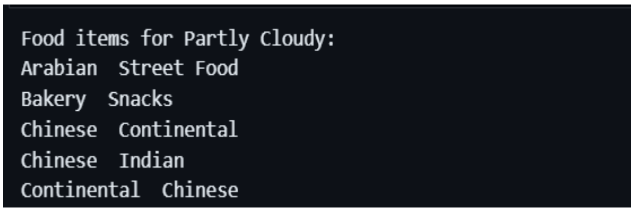

Fig. 13. Sample Food Categories for "Partly Cloudy"

Proceedings of 2024 International Conference on Cognitive Robotics and Intelligent Systems (ICC - ROBINS 2024)
IEEE Xplore Part Number: CFP24UD1-ART; ISBN: 979-8-3503-7274-8

Fig. 13 showcases the different food categories available. The further step for the user would be to enter the categories they serve in their restaurant considering a point that each restaurant or kitchen don't serve everything. Taken an example of the restaurant serving "Maharashtrian" & "Waffle", the recommendation based on those are presented in Fig. 14.

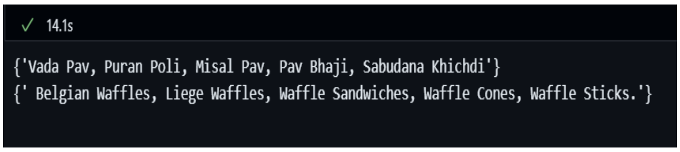

Fig. 14. Recommendations generated

The results from employing various machine learning algorithms for weather prediction and subsequent food recommendations demonstrated promising accuracy and efficiency. The models exhibited significant capabilities in forecasting weather patterns, enabling the system to make informed and adaptive food suggestions based on prevailing weather conditions. The removal of outliers enhanced the models' robustness, contributing to refined predictions. The evaluation of these algorithms revealed their diverse strengths, with each contributing uniquely to the overall accuracy and effectiveness of the system.

IV. Conclusion & Future Scope

With forecast of weather and integrating it with the food preferences, the proposed work reduced the food wastage and optimize inventory management. For prediction of weather 5 different algorithms were implemented to present a showcasing output with reduced errors and a promising result. However, as a future scope integration of additional meteorological data and exploring ensemble methods to enhance prediction can be done. Also adding up the system's future prospect can be on elevating the accuracy and user experience. Also adding up features like to adapt through customer trends and integration of other customer behavior can enhance the recommendation part. A graphical user interface for communication and utilization of these changes can enhance the end user's experience. In conclusion, while the present study marks a significant step toward addressing food wastage challenges in the cloud kitchen industry, continuous research and development remain essential to refine and optimize the system's accuracy, responsiveness, and applicability in real-world scenarios.

References

[1] Maria Carmela Annosi, Federica Brunetta, Francesco Bimbo, Marianthi Kostoula, Digitalization within food supply chains to prevent food waste. Drivers, barriers and collaboration practices, Industrial Marketing Management, Volume 93, 2021

[2] Zahir Irani, Amir M. Sharif, Habin Lee, Emel Aktas, Zeynep Topaloğlu, Tamara van't Wout, Samsul Huda, Managing food security through food waste and loss: Small data to big data, Computers & Operations Research, Volume 98, 2018

[3] Vinayak Bharadi, Pavan Jadhav, Omkar Nanche, Onkar Munj, Food Waste Management Using Machine Learning, 2022 IJCRT, Volume 10, Issue 4 April 2022, ISSN: 2320-2882

[4] H. Saima, J. Jaafar, S. Belhaouari and T. A. Jillani, "Intelligent methods for weather forecasting: A review," 2011 National Postgraduate Conference, Perak, Malaysia, 2011, pp. 1-6, doi: 10.1109/NatPC.2011.6136289.

[5] Choudhary, Nita. "Strategic analysis of cloud kitchen–a case study." *Management Today* (2019).

[6] Blum, David. "Ways to reduce restaurant industry food waste costs." *International Journal of Applied Management and Technology* 19.1 (2020): 1.

[7] Naik, Pratiksha Ashok. "Intelligent food recommendation system using machine learning." *Nutrition* 5.8 (2020).

[8] Saima, H., et al. "Intelligent methods for weather forecasting: A review." *2011 national postgraduate conference*. IEEE, 2011.

[9] O. Takahashi and S. Sasaki, "A Dinner Menu & Recipe Recommendation System with Food Expiration-date Notification Functions," 2022 13th International Congress on Advanced Applied Informatics Winter (IIAI-AAI-Winter), Phuket, Thailand, 2022, pp. 230-235, doi: 10.1109/IIAI-AAI-Winter58034.2022.00052.

[10] T. Mokdara, P. Pusawiro and J. Harnsomburana, "Personalized Food Recommendation Using Deep Neural Network," 2018 Seventh ICT International Student Project Conference (ICT-ISPC), Nakhonpathom, Thailand, 2018, pp. 1-4, doi: 10.1109/ICT-ISPC.2018.8523950.

[11] Rashmi, M.R., Vaithilingam, C.A., Kamalakkannan, S. and Narayanavaram, B., 2023, March. IoT for Agriculture System-Weather Prediction & Smart Irrigation System for Single Plot, Multiple Crops. In *2023 9th International Conference on Electrical Energy Systems (ICEES)* (pp. 536-541). IEEE.

[12] Susmitha, A., Alakananda, T., Apoorva, M.L. and Ramesh, T.K., 2017, August. Automated Irrigation System using Weather Prediction for Efficient Usage of Water Resources. In *IOP Conference Series: Materials Science and Engineering* (Vol. 225, No. 1, p. 012232). IOP Publishing.

[13] Baby, S.S. and Siji Rani, S., 2022, September. A Hybrid Product Recommendation System Based on Weather Analysis. In *Proceedings of the 6th International Conference on Advance Computing and Intelligent Engineering: ICACIE 2021* (pp. 263-273). Singapore: Springer Nature Singapore.

[14] Nair, L.S. and Cheriyan, J., 2023, January. Multi-Featured Movie Recommendation Using Knowledge Graph. In *2023 International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)* (pp. 590-594). IEEE.

[15] Priya, S.B., 2015. A survey on weather forecasting to predict rainfall using big data analytics. *IJISET-International J. Innov. Sci. Eng. Technol*, *2*(10), pp.879-886.

[16] Sharma, Urvi, G. P. Sajeev, and S. Siji Rani. "Personalized Fashion Recommendation Using Nearest Neighbor PageRank Algorithm." *2022 International Conference on Connected Systems & Intelligence (CSI)*. IEEE, 2022.

[17] Weather History https://www.wunderground.com/ (Accessed on December 2023)